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Can Interpersonal Trust Predict and Account for Symptom Change During Group Cognitive Processing Therapy for PTSD? An Investigation of the Iterated Trust Game

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Keywords: cognitive processing therapy | group psychotherapy | iterated trust game | PTSD | treatment mechanisms | treatment prediction

ABSTRACT

Objective: Group cognitive processing therapy (GCPT) is frequently utilized to treat PTSD within the VA healthcare system, but its mechanisms are not well understood. Interpersonal trust could be an important change process in GCPT given its relevance to group-based therapy and its role in CPT, but self-report measures are inadequate for capturing the dynamic interplay that defines interpersonal trust. Here, we examined the degree to which interpersonal could predict and account for PTSD symptom change in GCPT using the iterated trust game (ITG)—a behavioral task used to approximate real-world trust behavior.

Methods: Participants were Veterans with PTSD who participated in an effectiveness trial comparing a 12-week course of GCPT ($n = 37$) to a treatment-as-usual (TAU) waitlist condition ($n = 23$) of equivalent length. Both groups completed the ITG and measures of PTSD before and after treatment as well as a pencil-paper measure of interpersonal trust before treatment. Participants in GCPT completed measures of PTSD severity, group relationship quality, and therapist relationship quality at each treatment session.

Results: Pre-post changes in ITG-measured trust behavior did not differ between GCPT and TAU ($p = 0.075$). However, improvements in ITG scores partially accounted for decreased PTSD symptoms in GCPT, as demonstrated by a more modest change in PTSD symptoms when ITG was in, $b = -5.95$, $p = 0.032$, versus not in the model, $b = -9.05$, $p = 0.001$. Additionally,

Christopher Hunt and M. Wright Williams contributed equally to the work described here and share first authorship.

Brooks King-Casas and Pearl Chiu share senior authorship for the work described here.

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higher ITG scores, but not self-reported trust, predicted steeper reductions in PTSD symptoms, $b = -0.50$, $p = 0.042$, and improvements in group relationship quality, $b = 0.28$, $p = 0.037$, across GCPT sessions.

Conclusions: Interpersonal trust improvement may predict and account for symptom change in GCPT. Targeting interpersonal trust during GCPT could render the treatment more effective.

1 | Introduction

Posttraumatic stress disorder (PTSD) is a common and debilitating condition in Veterans (Maynard et al. 2017; Vogt et al. 2017; Scoglio et al. 2022). While several evidence-based psychotherapies have been developed to treat PTSD, they fail to produce clinically significant improvement in up to half of Veterans (Steenkamp et al. 2015, 2020). Research into PTSD treatment mechanisms has begun yielding useful insights for improving the effectiveness of PTSD therapies (Zalta 2015; Gallagher 2017; Cooper et al. 2017) but have been focused almost exclusively on individual formats. While the widespread proliferation of individual PTSD treatments makes this focus justified, research into the mechanisms of *group* PTSD treatments may also be useful for two reasons. First, group-based psychotherapies involve different relationship dynamics that may result in slightly different treatment mechanisms. For instance, success in group therapy depends not just upon developing a strong alliance with one's therapist but with members of one's therapy group as well (Burlingame et al. 2011). As a result, treatment mechanisms of individual psychotherapies may not apply to group-based formats, thereby necessitating studies of group-based mechanisms specifically (Coco et al. 2019; Yalom and Leszcz 2005). Second, improving the effectiveness of group treatments could increase their utilization in the Veterans Health Administration (VHA), which could help alleviate the long wait times at the VA by allowing several patients to be treated simultaneously (Author et al. 2017).

Among group PTSD treatments, group cognitive processing therapy (GCPT) is perhaps the most well-studied and utilized within the VA (Morland et al. 2011; Resick et al. 2015), making an understanding of its mechanisms particularly important. One theoretically important mechanism of GCPT is interpersonal trust, which is thought to play an essential role in group-based psychotherapies in general (Yalom and Leszcz 2005) and is an explicit focus of CPT specifically (Resick et al. 2017). The importance of addressing interpersonal trust in GCPT may be even more important for Veterans, who often have concerns about their capacity to trust family members, providers, civilians, and government institutions, including the VA itself (Haro et al. 2019). Moreover, lack of social support, which is frequently associated with poor interpersonal trust (Anderson and Dickinson 2010; Kim et al. 2012; Flores et al. 2014), was found be among the best predictors of PTSD severity in a large meta-analysis of Veterans (Blais et al. 2021), underscoring the contribution of interpersonal trust to the severity of PTSD.

Despite this theoretical framework, little work has been devoted to evaluating how interpersonal trust acts as a mechanism of GCPT or indeed any group-based PTSD treatment. One reason for this dearth of research is that interpersonal trust is a difficult concept to measure. Since interpersonal trust is a dynamic exchange between two or more people, self-report instruments—which are typically utilized to study psychotherapeutic mechanisms—have poor validity for measuring real-world trust behavior as they can only assess one person's perspective at a single point in time (Putnam 1995; Glaeser et al. 2000). Consequently, researchers, have increasingly turned to measuring trust through behavioral task paradigms like the iterated trust game (ITG). The ITG is a multi-round, social exchange game in which two people can play for mutual benefits through the expression and repayment of trust in the form of monetary rewards (King-Casas et al. 2005). In its most basic form, one player has the opportunity to "trust" their partner by investing a valued resource (e.g., monetary points) with them in the hope that their partner will repay that trust with a return on their investment. The value of the ITG is that it putatively mirrors patterns of many real-world trust interactions, which is supported by the ITG being able to sensitively measure trust behavior (Anderson and Dickinson 2010; Tzieropoulos 2013; Li et al. 2021, including among healthy populations (McCabe et al. 2001; Delgado et al. 2005) and in disorders characterized by relationship disruption (e.g., Autism spectrum disorder: Chiu et al. 2008; borderline personality disorder: King-Casas et al. 2008). Indeed, participants with PTSD have been found to offer significantly lower investments in the ITG both overall and when playing against a cooperative partner (Bell et al. 2019), underscoring the ability of the ITG to sensitively measure lower interpersonal trust capacity in this population.

Given both the theoretical importance of interpersonal trust for GCPT and the ability of the ITG to adequately capture it, the purpose of this study was to perform a preliminary examination into whether ITG-measured trust behavior can predict and account for symptom change during GCPT for Veterans with PTSD. We examined this question using data from a previously conducted effectiveness trial comparing GPCT to a treatment-as-usual (TAU) waitlist control condition in which participants in both conditions completed the ITG at pre- and posttreatment (Williams et al. 2022). We examined three distinct questions using this data set. First, we examined group differences in ITG-measured trust behavior from pre- to posttreatment. We hypothesized that changes in ITG trust would be greater in GCPT relative to TAU, consistent with GCPT enhancing trust behavior above and beyond what would be expected from other general treatments or from the passage of time. Second, we examined whether the effect of GCPT on PTSD symptoms was reduced when controlling for interpersonal trust. We hypothesized that the effect of GCPT (but not TAU) on PTSD symptoms would be rendered nonsignificant when accounting for changes in ITG trust behavior, consistent with change in interpersonal trust accounting for the effect of GCPT on PTSD symptoms. Finally, we examined whether ITG-measured trust behavior at pretreatment would predict changes in both PTSD symptoms and relationship quality with other group members and the patient's therapist across sessions of GCPT. We hypothesized

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that levels of pretreatment ITG trust behavior would predict steeper declines in PTSD symptoms and steeper improvements in group- and therapist- relationship quality across sessions. However, because better symptom change could theoretically be related to either lower interpersonal trust at baseline (i.e., allowing room for growth) or higher interpersonal trust at baseline (i.e., serving as a strength to spur symptom reduction; Hunt et al. 2024), this hypothesis was treated as exploratory. Together, these analyses should provide initial insight into whether interpersonal trust as measured by an objective cooperative exchange game acts can account for some of the effect of GCPT for Veterans with PTSD.

2 | Methods

Full details of methods can be found in the Supplement and only the most pertinent details will be presented here.

2.1 | Participants and Procedures

Participants were Veterans who sought treatment at PTSD clinics at a large VA Medical Center and who agreed to participate in a clinical trial of GCPT for versus TAU for PTSD (ClinicalTrials.Gov registration #NCT01244477). Inclusion criteria entailed: being between 18 and 50 years old and diagnosed with PTSD as established via a structured interview. Exclusion criteria included: dementia, a neurological disorder, active drug or alcohol abuse, contraindications to functional Magnetic Resonance Imaging (fMRI)[1], moderate to severe traumatic brain injury (TBI), active suicidality, or a concurrent diagnosis of a psychotic spectrum disorder. Since the original study included neuroimaging, subjects over 50 were excluded to minimize the impacts of brain atrophy that may begin accruing at this age (Ge et al. 2002).

Participants who met initial eligibility criteria completed a pretreatment assessment that included measures of PTSD, a self-report measure of interpersonal trust, and the ITG, which was completed in an fMRI scanner[1]. Following the pretreatment assessment, participants were given the option to initiate group CPT (GCPT) or treatment-as-usual (TAU)—which was a 12-week waitlist period equivalent to the length of GCPT. Although we originally planned to randomize treatment conditions, we ultimately allowed Veterans to choose their treatment condition (GCPT or TAU) to help mitigate a significant delay in initiating treatment that would have occurred for the TAU group, in-line with VA best-practice guidelines (U.S. Department of Veterans Affairs 2017; Mott et al. 2014). During CPT, participants completed weekly self-report assessments of PTSD symptoms and perceived relationship quality with their group and with their therapist. All participants (GCPT and TAU) completed a posttreatment assessment that included the same measures from pretreatment except for the self-report measure of interpersonal trust. TAU Veterans were offered GCPT immediately after completing TAU.

An initial sample of 87 Veterans were recruited for the study, of which $N = 61$ ($n = 38$ for GCPT, $n = 23$ for TAU) met all inclusion/exclusion criteria. For analyses of change within GCPT, only participants randomized to GCPT were necessarily included. One participant in the GCPT condition was excluded due to meeting the definition of an extreme outlier in terms of PTSD symptom change from pre- to posttreatment (Walfish 2006; Schwertman et al. 2004; see box plot in Supporting Information S1: Figure 1S). Since we utilized statistical analyses that were robust to partially missing data, all other participants who completed at least the baseline assessment were included, consistent with an intent-to-treat approach. Full information about participant flow and enrollment can be found in Figure 1 and full information on demographic and baseline clinical characteristics for the final sample can be found in Table 1. No differences in demographic or clinical variables between GCPT or TAU were observed at pretreatment (all $ps > 0.420$; see Table 1 for full set of GCPT-TAU comparisons).

All participants provided informed consent for this study. All recruitment, informed consent, and study components were conducted in compliance with the Institutional Review Board of the affiliated site, which approved the study.

2.2 | Treatment

GCPT was delivered according to the standard 12-session protocol used by Chard et al. (2009) to eight gender-homogeneous GCPT cohorts (six for males, two for females) that had 4–5

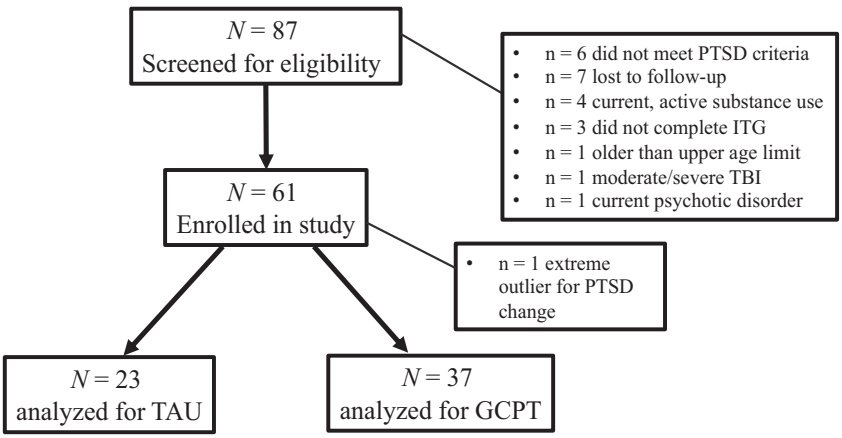

FIGURE 1 | Participant flow diagram. GCPT = Group CPT, ITG = Iterated Trust Game, TAU = Treatment as usual, TBI = Traumatic brain injury.

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TABLE 1 | Demographics and pretreatment clinical characteristics of the present sample.

	GCPT ($n = 37$)	TAU ($n = 23$)	t or χ^2
Gender			
Male (%)	31 (81.6)	19 (82.6)	0.10
Female (%)	7 (18.4)	4 (17.4)	—
Race/Ethnicity			
Non-Hispanic White (%)	15 (39.5)	12 (52.2)	1.96
Hispanic or Latino (%)	6 (15.8)	5 (21.7)	—
Black or African American (%)	14 (36.8)	5 (21.7)	—
Native American or Hawaiian (%)	1 (2.6)	0 (0)	—
Unknown Race/Not Reported (%)	2 (5.3)	1 (4.3)	—
Educational Attainment			
Some High School (%)	1 (2.6)	0 (0)	1.88
High School/GED (%)	6 (15.8)	4 (17.4)	—
Some College (%)	23 (60.5)	12 (52.2)	—
Bachelor's Degree (%)	5 (13.2)	6 (26.1)	—
Some Graduate School or Advanced Degree (%)	3 (7.9)	1 (4.3)	—
Marital Status			
Married (%)	16 (42.1)	12 (52.2)	1.19
Single (%)	6 (15.8)	2 (8.7)	—
Divorced (%)	5 (13.2)	2 (8.7)	—
Long-Term Dating (%)	6 (15.8)	4 (17.4)	—
Other (%)	4 (10.5)	2 (8.7)	—
Service Era Joined			
Post-Vietnam (%)	5 (13.2)	1 (4.3)	1.35
Persian Gulf War (%)	18 (47.4)	12 (52.2)	—
OEF/OIF/OND (%)	14 (36.8)	9 (39.1)	—
Unknown/Not Reported (%)	1 (2.6)	1 (4.3)	—
PTSD Symptoms			
Mean PCL-M (SD)	62.50 (11.63)	59.96 (12.21)	0.81
Mean CAPS-IV (SD)	71.66 (23.28)	67.52 (24.71)	0.66
Trust Measures			
Mean RITS (SD)	58.54 (11.38)	58.00 (10.17)	0.19
Mean ITG IR (SD)	0.29 (0.15)	0.32 (0.23)	−0.75

Note: Differences in count variables were assessed with chi-square difference tests. Differences in continuous variables were assessed with *t*-tests. All comparisons were nonsignificant ($p > 0.05$).
Abbreviations: CAPS-IV, Clinician Administered PTSD Scale for DSM-IV; GCPT, Group cognitive processing therapy; GED, General Education Degree; IR, Investment Ratio; OEF, Operation Enduring Freedom; OIF, Operation Iraqi Freedom; OND, Operation New Dawn; PCL-M, PTSD Checklist Military Version; RITS, Rotter Interpersonal Trust Scale; TAU, Treatment as usual.

Veterans per group. We opted against mixed-gender groups since this might discourage disclosure and discussion of male-perpetrated traumas among female Veterans, such as military sexual trauma. All three group therapists were certified VA CPT providers. Sessions were independently evaluated and exhibited very high fidelity to the treatment manual (see Supplement for more details). Veterans in the TAU condition were asked to continue attending their usual VA services, including medication management when applicable, and did not receive psychotherapy with the study.

2.3 | Measures

2.3.1 | Clinical Measures

Upon enrollment, PTSD diagnoses were established using the Clinician Administered PTSD Scale for DSM-IV (CAPS-IV; Blake et al. 1995), which is a gold-standard semi-structured interview for PTSD. PTSD severity levels were also tracked before and after treatment (both groups) and at each treatment session (GCPT only), using the PTSD Checklist Military Version

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(PCL-M; Weathers et al. 1993), which served as our main PTSD outcome. Participants also completed the Group Evaluation Scale (GES; Hess 1996) and the Working Alliance Inventory – Short Form (WAI-SR; Hatcher and Gillaspy 2006) at each GCPT session, which served as within-treatment measures of perceived relationship quality with one's therapy group and one's therapist, respectively. The GES is a brief, 7-item self-report measure of a patient's perceived benefit of being part of a therapy group while the WAI-SR is a brief self-report measure of a patient's perceived therapeutic alliance with their therapist. Since the present study was concerned primarily about trust and relationship quality, we only examined the bond subscale of the WAI-SR (WAI-SR Bond), which reflects the degree of trust and likability the patient has for their therapist. Finally, participants completed the Rotter Interpersonal Trust Scale (RITS) before treatment, which is a 25-item self-report measure of interpersonal trust. The RITS served as a pencil-paper measure of interpersonal trust that could be compared to the ITG in terms of predictive capabilities.

2.3.2 | The Iterated Trust Game (ITG; King-Casas et al. 2005)

The ITG is a computerized social exchange game in which two people can mutually benefit from the expression and repayment of trust. In the game, one player serves as the "investor" and the other as the "trustee." The investor begins with $20, and the trustee begins with no money. On each round, the investor chooses to "invest" a certain amount of money that is sent to the trustee, which triples in value upon transfer. The trustee then has the option to send a portion of this money back to the investor, ranging from 0% to 100%. The primary outcome measure from the ITG was the investment ratio (IR), which is defined as the percent of the participant's total money that they sent to the trustee, averaged across all 10 rounds of the ITG. Higher IR scores reflect greater interpersonal trust behavior.

Before beginning the ITG, participants were told that they would either be playing against a computer or a real person who was connected to the game via the internet. However, all participants played against a computer algorithm that exercised the same strategy across participant so that experimental conditions were standardized and so that scores could be more readily compared and interpreted. Participants earned $20–$40 for playing the ITG, with higher total scores equating to greater monetary earnings. Participants were made aware of this contingency before beginning the ITG. The ITG takes approximately 10 min to complete, not including time required to enter and exit the fMRI scanner.

2.4 | Data Analysis

Pearson correlations were conducted to examine relations between measures of trust (ITG IR, RITS) and PTSD (CAPS-IV, PCL-M). All subsequent analyses were conducted using mixed effects models (MEMs), which are ideally suited for analyzing longitudinal change given their ability to account for individual variability across time and accommodate partially missing data (Curran et al. 2010). Two different types of MEMs were tested: Models of pre-post change, which included participants from both treatment groups, and models of session-by-session change, which included only participants from GCPT (i.e., there was no session-by-session data for TAU). For both model types, we first established the best-fitting unconditional model of change by comparing fit indices of models utilizing different combinations of random intercepts and slopes (see Supplement for more details). Effect sizes are reported as unstandardized *b* values. All models were estimated using maximum likelihood estimation procedures. Time variables and the treatment group variable were zero-centered, and all other predictors were standardized to reduce multicollinearity (Aiken et al. 1991).

2.4.1 | Analyses of Pre-to-Post Treatment Changes

The first set of MEMs were models of pre-post treatment changes. The purpose of these models was to test whether changes in interpersonal trust behavior (IR scores) and PTSD symptoms (PCL-M scores) from pre-to-post treatment differed between treatment groups (GCPT and TAU). For each outcome (IR scores, PCL-M scores), we tested an MEM that included time, treatment group, and the Time x Treatment group interaction as fixed effects. The Time x Treatment group effect tested the hypothesis that improvements in trust behavior and PTSD symptoms would be greater in GCPT relative to TAU. The *b* value for the interaction reflects the difference between treatment groups in IR/PCL-5 score change from pre-to-post treatment. For interpretative purposes, we followed up this analysis by examining the marginal effects time on each outcome separately for each treatment group.

Next, to test whether changes in PTSD symptoms might be attributable to changes in interpersonal trust behavior, we re-tested the PCL-M MEM with IR scores included as a time-varying covariate. Here, we hypothesized that significant reduction in PCL-M scores from pre- to posttreatment in the GCPT group would be rendered nonsignificant when including IR scores in the model. These MEMs were followed up with Pearson correlations of residualized change scores for IR and PCL-M (see Supplement for calculation) for each group (GCPT, TAU) to illustrate the degree to which changes in interpersonal trust were associated with changes in PTSD symptoms in each treatment group.

2.4.2 | Analyses of GCPT Session-by-Session Changes

The next set of MEMs modeled changes in variables that were collected intensively across GCPT sessions—namely PTSD symptoms (PCL-M), group relationship quality (GES), and therapist relationship quality (WAI-SR Bond). As such, session served as the time variable in these models. For PCL-M, time was a 13-level within-subjects variable corresponding to baseline plus the 12 sessions of GCPT. Since data on GES or WAI-SR Bond necessarily could not be collected before treatment (i.e., they are based on ratings of one's group and one's therapist), time was only a 12-level within-subjects corresponding to the 12 sessions of GCPT. Models of each outcome (PCL-M, GES, WAI-SR Bond), fixed effects included pretreatment IR, session, and

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the Pretreatment IR x Session interaction. The IR x Session interaction tested whether the slope of change in PCL-M, GES, or WAI-SR Bond across GCPT was related to levels of interpersonal trust behavior on the ITG at pretreatment. We then repeated these same analyses with baseline RITS scores to compare effects of baseline IR to those obtained from a pencil-paper measure of interpersonal trust. For GCPT session models, the *b* value of the interaction reflects the change in outcome (PCL-M, GES, or WAI-SR Bond) per session associated with a one unit increase in the baseline trust measure (IR or RITS). For models of GES or WAI-SR Bond, baseline PCL-M was included as a covariate to control for initial levels of symptom severity. For PCL-M models, baseline PCL-M was already accounted for by time 0 in the model.

Data was analyzed using SPSS version 28. Alpha was set at 0.05 (two-tailed) for all statistical tests.

3 | Results

3.1 | Associations Between Study Variables at Baseline

PCL-M scores correlated strongly and positively with CAPS-IV total scores, $r(60) = 0.61$, $p < 0.001$, neither of which correlated significantly with either trust measure ($ps > 0.069$). Surprisingly, RITS and IR scores did not significantly correlate with each other either, $r(60) = 0.12$, $p = 0.369$, meaning that self-reported trust was unrelated to trust as measured in the ITG.

3.2 | Model Fitting

For models of pre-post change, a model that included a random intercept slope afforded the best balance of fit and parsimony while the correlated slope model provided the best fit for the models of session-by-session change in GCPT. More details of model fitting results can be found in Table 1S of the Supplement.

3.3 | Changes in Interpersonal Trust and PTSD Symptoms From Pre-to-Post Treatment

MEMs of pre-post changes revealed a trending but non-significant Timepoint x Treatment group interaction for predicting both IR scores, $b = -0.11$, 95% CI = −0.22, 0.011, $t = -1.82$, $p = 0.075$, and PCL-M scores, $b = 8.67$, 95% CI = −1.40, 18.73, $t = 1.71$, $p = 0.091$. Follow-up analyses of pre-post changes for each treatment group indicated that that trend-level interaction for IR was driven by IR scores increasing in the GCPT group, $b = 0.054$, 95% CI = −0.017, 0.12, $t = 1.54$, $p = 0.133$, and decreasing in TAU, $b = -0.053$, 95% CI = −0.16, 0.049, $t = -1.08$, $p = 0.292$, although neither change reached statistical significance. For PCL-M changes, follow-up analyses indicated that the trend-level interaction was driven by PCL-M scores decreasing significantly in the GCPT group, $b = -9.05$, 95% CI = −14.30, −3.79, $t = -3.49$, $p = 0.001$, but not TAU, $b = -0.07$, 95% CI = −3.18, 3.04, $t = -0.05$, $p = 0.963$. Fitted value plots illustrating pre-post changes in IR and PCL-M for each treatment group can be found in Figure 2.

Next, to test whether the significant reductions in PTSD Symptoms for GCPT were partially attributable to changes in interpersonal trust, we reran the PCL-M MEM for the GCPT group entering IR as a time-varying covariate. With IR in the model, the effect of time was about 33% lower, $b = -5.95$, 95% CI = −11.36, −0.56, $t = -2.24$, $p = 0.032$, though it remained significant. Thus, improvements in interpersonal trust behavior appeared to explain some of the PTSD symptom reduction in GCPT, though a significant degree of reduction still existed independent of interpersonal trust changes.

To further illustrate links between changes in interpersonal trust and PTSD symptoms, we tested associations between residualized changes in IR and PCL-M from pre- to posttreatment for each treatment group. Consistent with findings from the MEM, residualized change in IR was found to be negatively associated with residualized changes in PCL-M, $r(25) = -0.56$, $p = 0.004$, indicating that participants in GCPT who showed

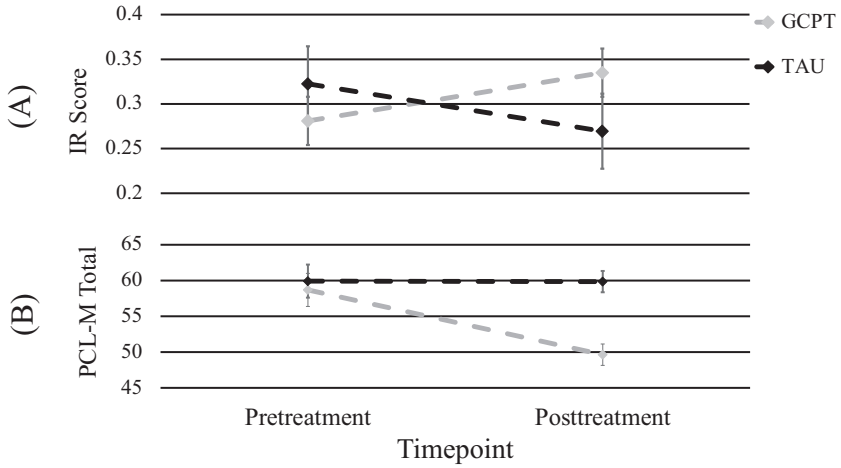

FIGURE 2 | Fitted value plot of changes in (A) IR scores and (B) PCL-M scores from pre- to posttreatment among participants enrolled in GCPT (gray) versus TAU (black). IR scores were collected in the ITG before and after treatment (GCPT) or an equivalent waitlist period (TAU). Error bars reflect the standard error of intercept (pretreatment) and slope (posttreatment) for each group at each timepoint. GCPT = Group cognitive processing therapy, IR = Investment ratio, ITG = Iterated Trust Game, PCL-M = PTSD Checklist Military Version, TAU = Treatment as usual.

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greater improvement in ITG-measured trust behavior tended to show greater declines in PTSD symptoms from pre- to post-treatment. In contrast, residualized changes in IR were not associated with residualized pre- to posttreatment PCL-M scores for TAU participants, $r(20) = -0.11$, $p = 0.661$ Scatterplots depicting the relationship between pre-post changes in IR and PTSD symptoms for each group (GCPT and TAU) can be found in Figure 3.

3.4 | Interpersonal Trust as a Predictor of GCPT Response

Next, we next tested whether changes in PCL-M, GES, and WAI-SR Bond scores across GCPT sessions could be predicted from pretreatment IR and RITS scores. Results of these models are in Table 2. These tests revealed that the Session x Pretreatment IR interaction was a significant predictor of changes in PCL-M, $b = -0.47$, $p = 0.039$, and GES, $b = 0.28$, $p = 0.037$, across sessions of GCPT. Thus, higher interpersonal trust behavior on the ITG at pretreatment predicted a steeper decline in PTSD symptoms and a steeper improvement in perceived relationship quality with other group members across sessions of GCPT. To illustrate these effects, we constructed fitted value plots depicting changes in PCL-M and GES scores across GCPT sessions at low (−1SD) and high (+1 SD) levels of pretreatment IR. These plots can be found in Figure 4.

In contrast, the RITS x Session interaction failed to predict changes in PCL-M scores ($p = 0.970$) or GES scores ($p = 0.125$) during GCPT. Thus, prediction of PTSD symptom reduction and group relationship improvement was achieved for the ITG but not a pencil-paper measure of interpersonal trust. Additionally, changes in the WAI-SR Bond subscale were not predicted by the Pretreatment IR x Session interaction or the Pretreatment RITS x Session interaction ($ps > 0.299$). Thus, changes in participants' perceived relationship quality with their therapist was unrelated to pretreatment levels of interpersonal trust capacity as measured behaviorally or through self-report.

4 | Discussion

The present study was a preliminary investigation into whether interpersonal trust acts can predict and account for symptom change during group cognitive processing therapy (GCPT) for Veterans with PTSD. A unique aspect of this study was that interpersonal trust was measured using a cooperative exchange game (the iterated trust game; ITG), which was administered at both pre- and posttreatment to Veterans who participated in GCPT and a treatment-as-usual (TAU) control condition. Contrary to expectations, changes in ITG-measured interpersonal trust behavior from pre- to posttreatment did not significantly differ between GCPT and TAU. However, pre-to-post improvements in interpersonal trust appeared to partially account for reduced PTSD symptoms in the GCPT group, implicating interpersonal trust as a potential change process of GCPT. Moreover, better interpersonal trust on the ITG at pretreatment predicted both greater declines in PTSD symptoms and greater improvements in relationship quality with other group members across sessions of GCPT, and these same predictive effects were not observed for a pencil-paper measure of interpersonal trust. Together, these findings suggest that while GCPT may not be optimized to shift interpersonal trust behavior, interpersonal trust may nonetheless be related to symptom change during GCPT when measured in an objective gamified fashion.

That declines in PTSD symptoms during GCPT were lessened when accounting for changes in interpersonal trust suggests that some of the effect of GCPT on PTSD symptoms might be related to changes in interpersonal trust. Evidence for this assertion is further supported by three other pieces of evidence. First, the fact that this correlation between pre-post changes in ITG trust and PTSD symptoms was not also observed in the TAU condition suggests that any reduction in PTSD associated with interpersonal trust improvement was driven by the procedures of GCPT as opposed to more general changes that would have been shared with TAU (e.g., regression to the mean because of the passage of time; Hsu 1995). Second the fact that interpersonal trust was measured via task rather than self-

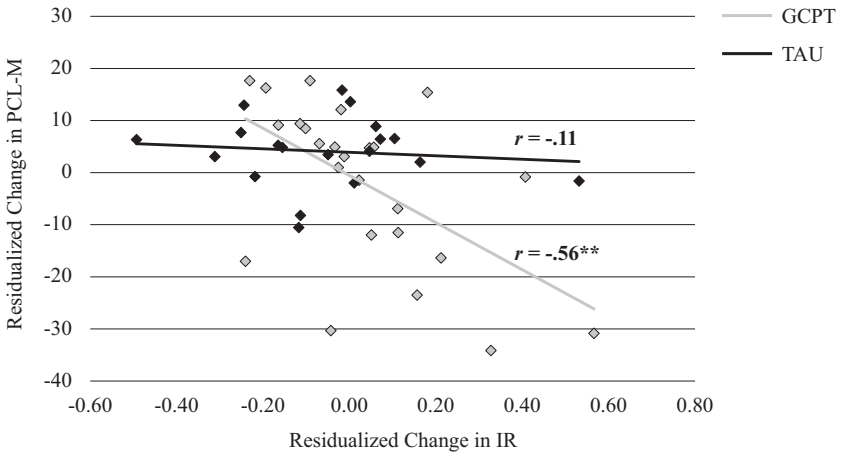

FIGURE 3 | Associations between residualized changes in IR and PCL-M from pre- and posttreatment for the GCPT group (gray) and TAU group (black). Residualized change scores were obtained by regressing the posttreatment measure (IR or PCL-M) onto its corresponding pretreatment measure and saving the residuals; higher scores represent a greater increase from pre- to posttreatment. IR = Investment ratio, PCL-M = PTSD Checklist for DSM-IV Military Version, GCPT = Group cognitive processing therapy, TAU = Treatment as usual. $^{**}p < 0.01$.

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TABLE 2 | Mixed effects model predicting changes in PTSD symptoms and relationship variables across GCPT sessions from baseline trust measures.

Model	Dependent variable	Predictor	b (95% CI)	*t*	p
1	PCL-M	Pre IR	−6.24 (−9.81, −2.68)	−3.57	0.001
		Session	−0.97 (−1.47, −0.47)	−3.96	< 0.001
		Pre IR * Session	−0.47 (−0.92, −0.03)	−2.17	0.039
2	GES	Pre IR	0.68 (−1.57, 2.94)	0.62	0.542
		Session	0.62 (0.34, 0.89)	4.69	< 0.001
		Pre IR * Session	0.27 (0.03, 0.52)	2.30	0.028
3	WAI-SR Bond	Pre IR	−0.09 (−1.35, 1.17)	−0.15	0.882
		Session	−0.16 (−0.28, −0.03)	−2.78	0.017
		Pre IR * Session	0.06 (−0.06 – 0.17)	1.08	0.302
4	PCL-M	Pre RITS	0.16 (−4.60, 4.93)	0.07	0.945
		Session	−0.97 (−1.50, −0.45)	−3.81	< 0.001
		Pre RITS * Session	0.01 (−0.45, 0.65)	0.37	0.716
5	GES	Pre RITS	−0.62 (−3.15, 1.92)	−0.50	0.624
		Session	0.63 (0.34, 0.93)	4.43	< 0.001
		Pre RITS * Session	−0.24 (−0.55, 0.06)	−1.64	0.115
6	WAI-SR Bond	Pre RITS	−0.09 (−1.32, 1.14)	−0.14	0.886
		Session	−0.13 (−0.23, −0.02)	−2.34	0.020
		Pre RITS * Session	0.04 (−0.07, 0.15)	0.70	0.488

Note: All predictors were tested in a base model that included a random intercept, random slope and a random intercept-slope correlation. In PCL-M models, session was a 13-level within-subjects variable that corresponded baseline plus each session of GCPT. Session was a 12-level variable in other models and baseline PCL-M was included as a covariate. Session was zero-centered; RITS and IR were z-scored. PCL-M, GES, and WAI-SR Bond were collected during each weekly session of GCPT.
Abbreviations: CI, Confidence interval; GCPT, Group cognitive processing therapy; GES, Group Evaluation Scale; IR, Investment ratio; Pre, Pretreatment; PCL-M, PTSD Checklist for DSM-IV Military Version; RITS, Rotter Interpersonal Trust Scale; WAI-SR Bond, Working Alliance Inventory Scale Revised Bond subscale.

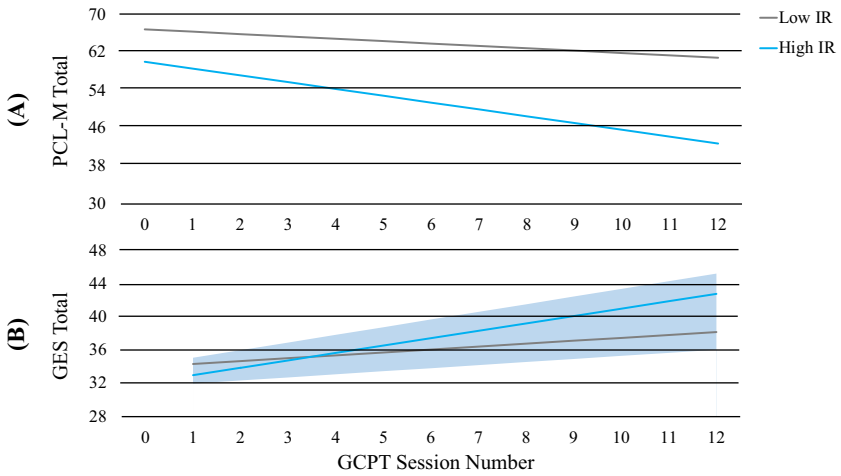

FIGURE 4 | Fitted value plots of changes in (A) PCL-M scores and (B) GES scores across GCPT sessions at levels of low (−1 SD; gray lines) and high (+1 SD; blue lines) IR scores at pretreatment as assessed during the ITG. Session zero refers to the baseline assessment; session 1–12 refers to the GCPT session number. Shaded areas indicate the 95% confidence interval for the effect. GCPT = Group Cognitive Processing Therapy, GES = Group Environment Scale, IR = Investment ratio, ITG = Iterated Trust Game, PCL-M = PTSD Checklist Military version.

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report helps rule out the possibility that its association with PTSD change was merely driven by shared method variance, which can augment associations among self-report measures (Campbell and Fiske 1959; Kenny 2014). Finally, the fact that pretreatment ITG-measured trust behavior predicted changes in PTSD across sessions suggests that trust behavior drove changes in PTSD symptoms and was not merely a correlate of PTSD change (Kraemer 1997). For instance, Veterans who began with a greater capacity for interpersonal trust may have been more amenable to restructuring their stuck points surrounding themes of interpersonal trust, which are addressed as part of CPT (Resick et al. 2017). Alternatively, the faster improvements in group relationship quality associated with higher ITG trust behavior may have been partially responsible for the more rapid reduction in PTSD symptoms (Yalom and Leszcz 2005).

Interestingly, higher pretreatment trust behavior predicted improvements in perceived relationship quality with other group members but not with one's therapist. One explanation for this discrepancy is that trust behavior on the ITG des not adequately capture the type of trust that develops between patient and therapist. Participants believed they were possibly playing against other participants during the ITG, making the task is putatively better equipped to measure trust between peers, such as therapy group members, as opposed to trust between leader and subordinate, such as patient and therapist. Indeed, research into trust within other areas (e.g., employment), suggest that so-called horizontal trust (i.e., peer-to-peer) and vertical trust (i.e., employee-to-manager) are not only distinct, but that the latter is more complex owing to the power differential that affects issues such as resource allocation (Krot and Lewicka 2012). The increased complexity of vertical trust may also explain why it was difficult to predict changes in patient-therapist trust from baseline ITG scores alone.

Unexpectedly, GCPT was not associated with greater improvements in interpersonal trust behavior compared to TAU. Combined with the above findings, this result suggests that improving interpersonal trust might help reduce PTSD symptoms during GCPT, but GCPT is currently not optimized to address interpersonal trust. Notably, the difference in interpersonal trust improvement between GCPT and TAU was trending toward significance, suggesting that GCPT is possibly capable of improving interpersonal trust, but only to a modest degree

Our results must be considered in light of several important limitations. First, because our study lacked an active control treatment, we cannot speak to whether improved interpersonal trust behavior is a unique mechanism of GCPT or one shared with other treatment protocols. Second, we had a relatively small sample for analyses of pre-post data and our participants were fairly homogeneous in terms of gender (mostly male) and age (under 50). Both factors limit the generalizability of our findings, as small samples do not allow for detection of more subtle differences in treatment response (e.g., latent treatment response groups) and because Veterans of different demographic backgrounds may exhibit different trust dynamics. Third, we did not measure ITG trust intensively during treatment, which prevented us from employing more complex statistical techniques (e.g., cross-lagged panel modeling) that are useful for more confidently disentangling directionality (i.e., whether changes in ITG trust predict changes in PTSD symptoms or the opposite). Fourth, we did not assess the degree to which participants believed they were playing against a real person during the ITG, which could impact the ability of their scores to adequately capture interpersonal trust. Fifth, we did not include the self-reported measure of trust (i.e., the RITS; Rotter 1967) at posttreatment, which necessarily prevented us from comparing the degree to which GCPT shifted trust in the ITG versus self-reported trust. Finally, participants in our study were not randomized, but rather self-selected into their condition of choice (GCPT or TAU). While allowing participants to select their treatment condition more accurately mirrors a real-world clinical setting, the lack of randomization does introduce potential third variables into our design since participants who self-selected into GCPT could be categorically different than those who opted for TAU. Notably, no differences in symptoms or demographic characteristics were found between the GCPT and TAU conditions at pretreatment, providing some assurance that the detrimental effect of this design flaw was minimized.

Clinically, our findings suggest that modifying GCPT to target interpersonal trust more directly could render the treatment more effective. Defining these modifications lies outside the scope of this paper, but group-based PTSD treatments that more explicitly address trust issues such as interpersonal therapy (IPT) (Robertson et al. 2004; Ray and Webster 2010) could provide useful templates for how to implement this change. Relatedly, a more modest change could be to shift the trust module (currently in session 9) to earlier in the protocol, so that patients are given greater time to contemplate and address stuck points related to trust during the treatment. A second implication is that pretreatment levels of interpersonal trust could eventually be used to help identify candidates who will be successful in GCPT or who should be targeted for interpersonal improvement with additional pretreatment sessions. Notably, the fact that the ITG is computerized, relatively brief (10 min when delivered outside the scanner), and purely behavioral increases the likelihood that it could be implemented as a measure of interpersonal trust in clinical settings.

4.1 | Conclusions

The present study provided preliminary evidence that interpersonal trust, as measured through a gamified behavioral task, may account for and predict symptom change during GCPT for Veterans with PTSD. Future studies should expand upon our findings by testing interpersonal trust has a similar role in other treatment protocols, replicating our study in a more demographically diverse sample, sampling interpersonal trust intensively during treatment to more confidently evaluate directionality, and utilizing a randomized control trial design. Ultimately, this study may allow interpersonal trust to function as a novel treatment target that can help augment the efficacy of GCPT.

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Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

[1] As a result of an acquisition parameter error by the fMRI technician, fMRI data were rendered unusable and is therefore not reported.

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